

□ TITANIC DATASET

Exploratory Data Analysis Report

Task 5 | Python · Pandas · Matplotlib · Seaborn

- 891 Training Passengers
- 12 Features
- 38.4% Overall Survival Rate
- 61.6% Did Not Survive
- 3 Passenger Classes
- 2 Sexes

Objective: Extract insights using visual and statistical exploration

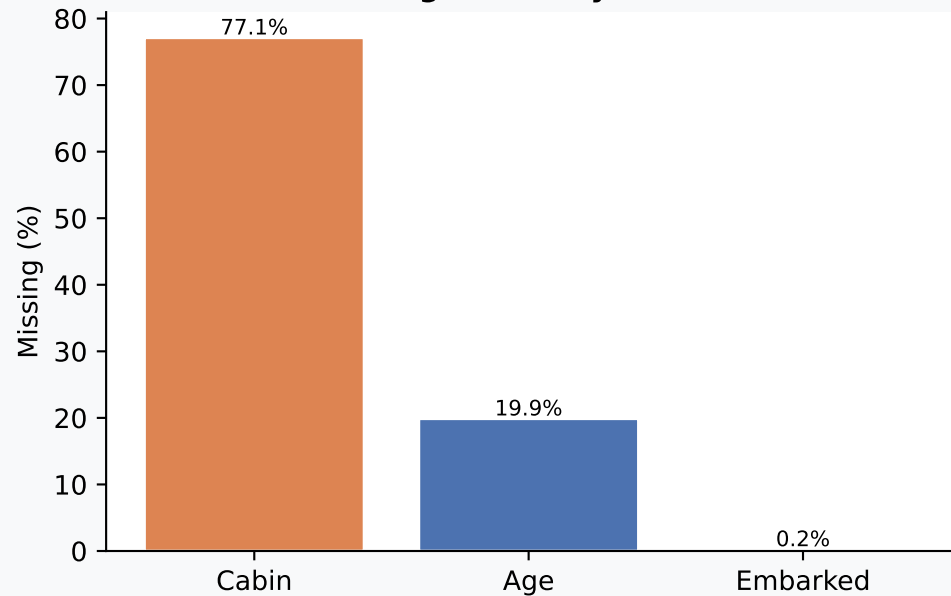
SECTION 1: Dataset Overview & Data Quality

Understanding the structure and completeness of our data

Dataset at a Glance

Dataset:	Titanic Passenger Survival
Training Rows:	891 passengers
Test Rows:	418 passengers
Target Variable:	Survived (0=Died, 1=Lived)
Numeric Features:	Age, Fare, SibSp, Parch
Categorical Features:	Sex, Pclass, Embarked
High Missing:	Cabin (77%), Age (20%)

Missing Values by Column



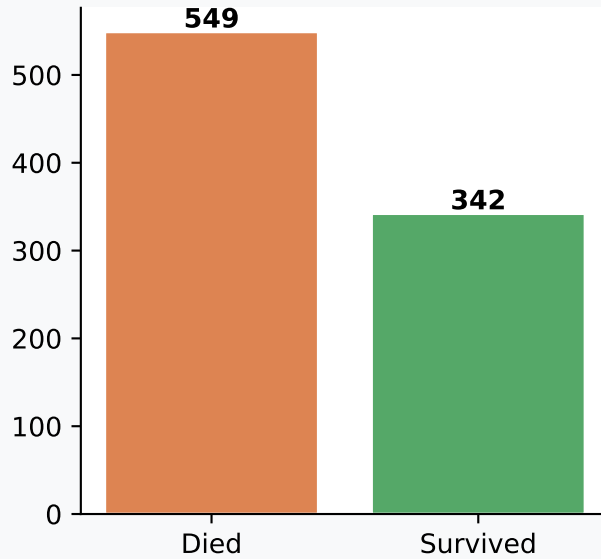
Statistical Summary (.describe())

	Age	Fare	SibSp	Parch	Survived	Pclass
count	714.0	891.0	891.0	891.0	891.0	891.0
mean	29.7	32.2	0.52	0.38	0.38	2.31
std	14.53	49.69	1.1	0.81	0.49	0.84
min	0.42	0.0	0.0	0.0	0.0	1.0
25%	20.12	7.91	0.0	0.0	0.0	2.0
50%	28.0	14.45	0.0	0.0	0.0	3.0
75%	38.0	31.0	1.0	0.0	1.0	3.0
max	80.0	512.33	8.0	6.0	1.0	3.0

SECTION 2: Survival Overview & Gender Analysis

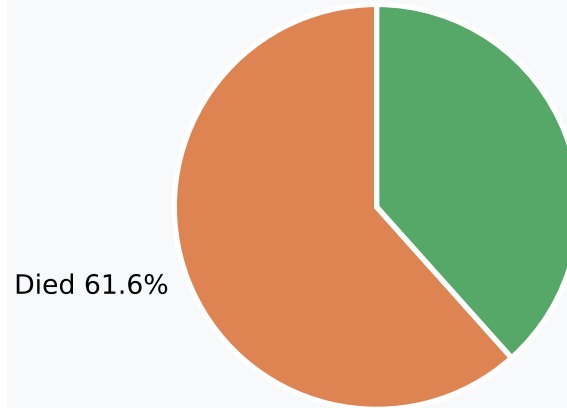
The most powerful predictor: Sex

Survival Count

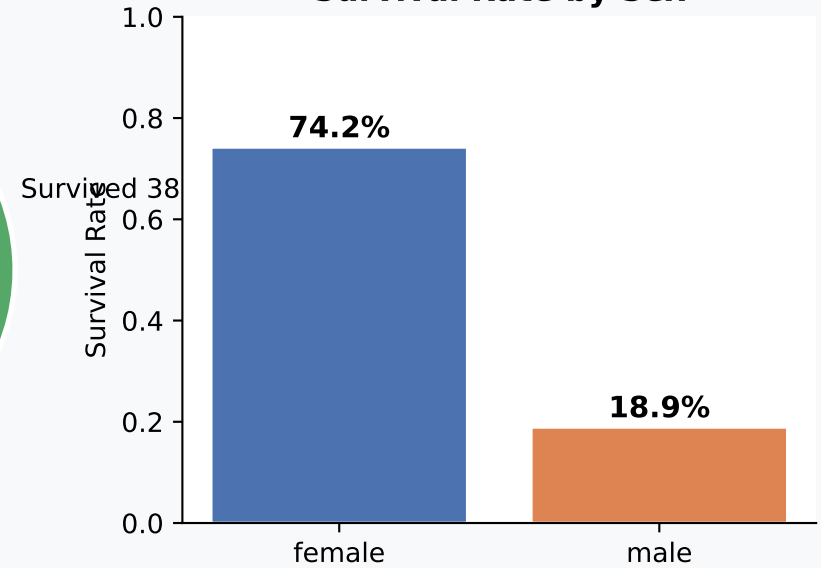


△ Only 38.4% survived.
A majority (61.6%) perished.

Survival Proportion

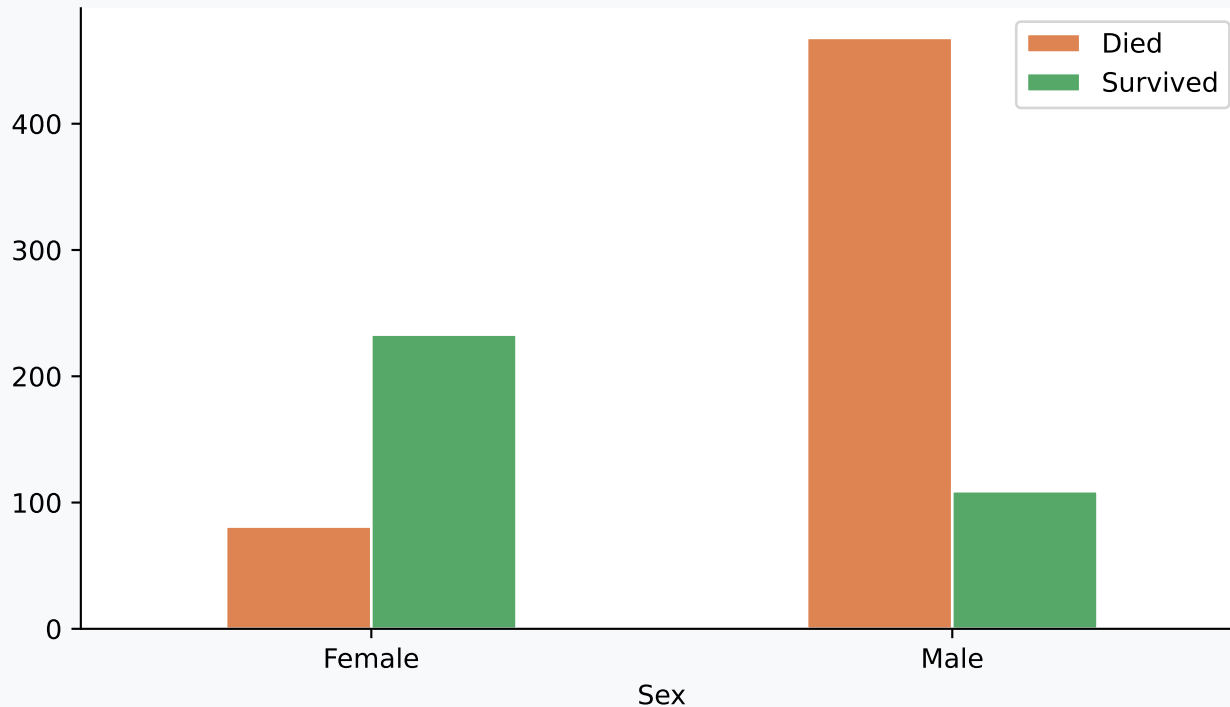


Survival Rate by Sex



□ Female: 74.2% vs Male: 18.9%
Strongest single predictor!

Survival Count by Sex



Key Observations:

- Female survival rate: 74.2%
- Male survival rate: 18.9%
- ~4× difference in odds

- Reflects "women and children first" protocol

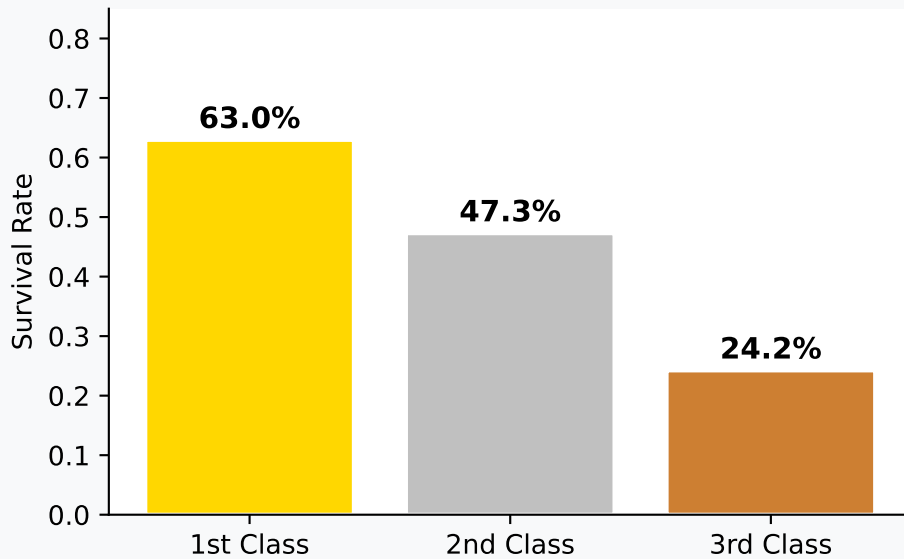
- Of 314 females: 233 survived
- Of 577 males: 109 survived

→ **Sex is the #1 most important**
feature for survival prediction

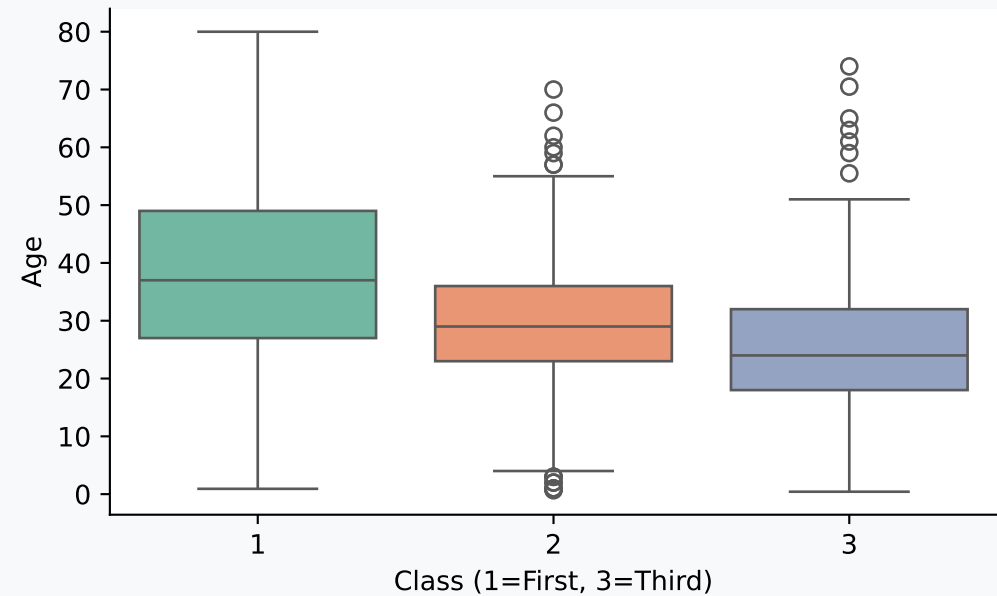
SECTION 3: Passenger Class & Age Analysis

Socioeconomic inequality and age effects on survival

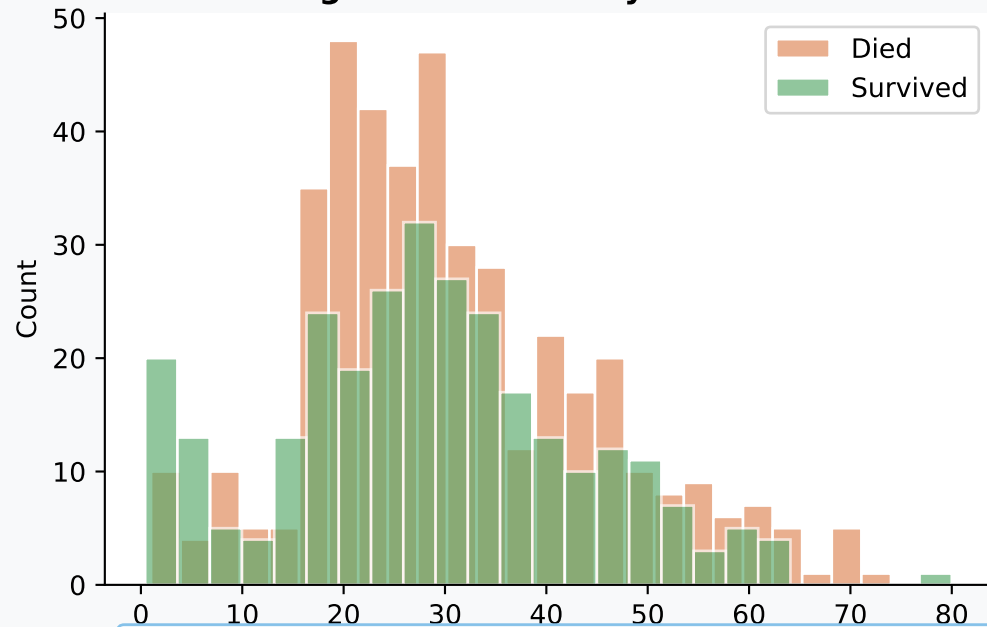
Survival Rate by Passenger Class



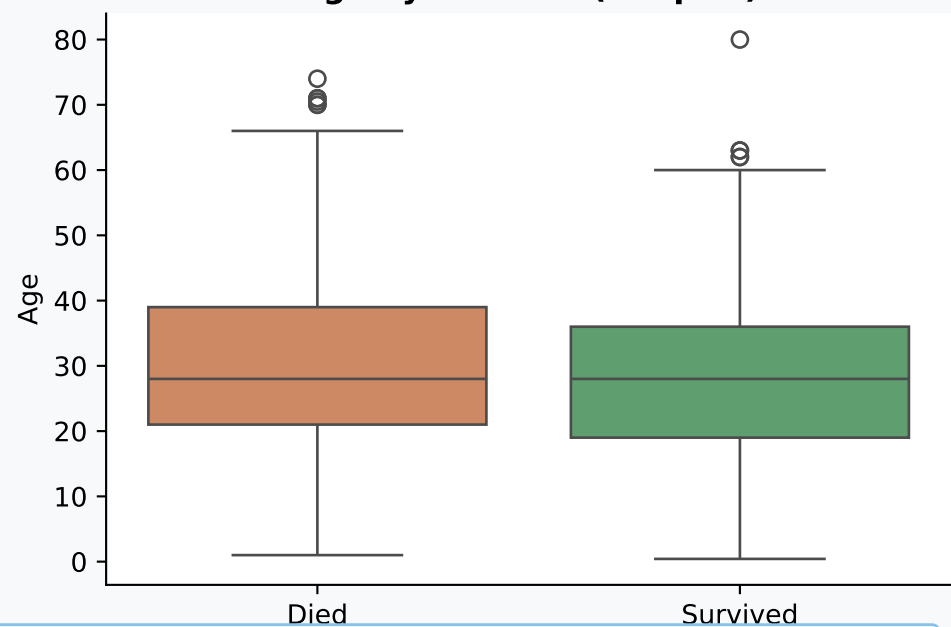
Age Distribution by Class



Age Distribution by Survival



Age by Survival (Boxplot)

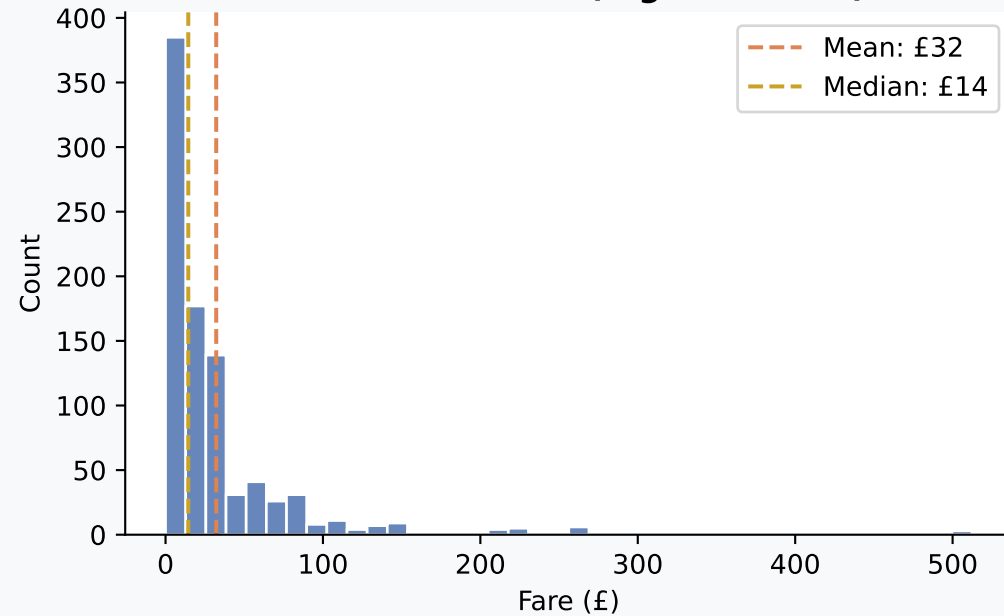


Observations: 1st Class: 63% survived | 2nd: 47% | 3rd: 24%. Median age ~28. Children had slightly better survival odds. Age alone is a weak predictor.

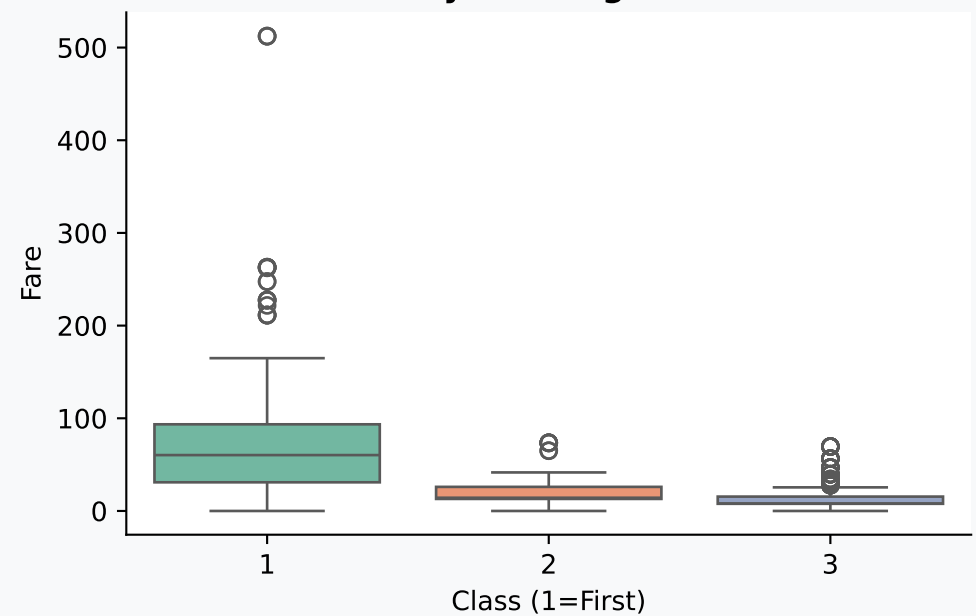
SECTION 4: Fare Analysis & Correlation Heatmap

How ticket prices relate to class and survival

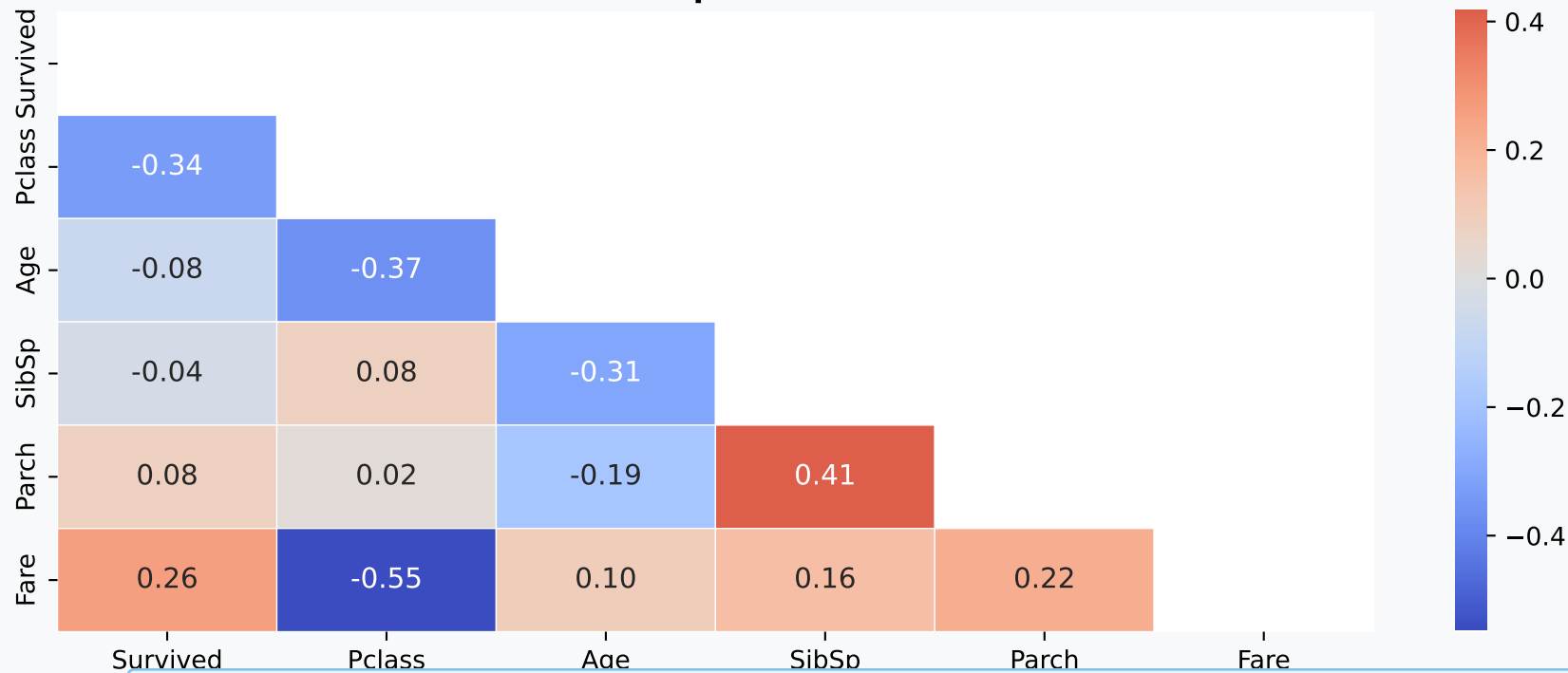
Fare Distribution (Right-Skewed)



Fare by Passenger Class



Correlation Heatmap of Numeric Features

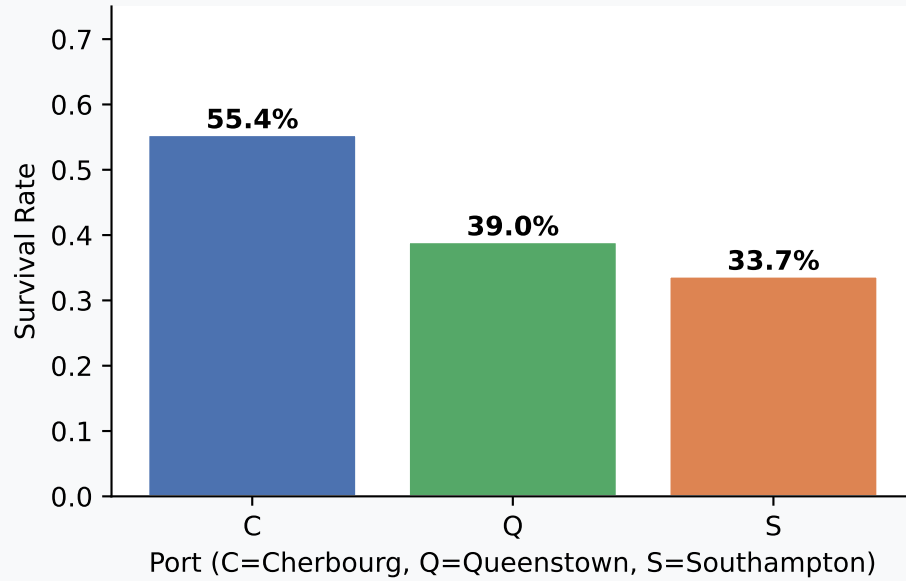


□ Fare (+0.26) & Pclass (-0.34) are top correlates of Survived. Pclass and Fare are strongly inversely correlated (-0.55) — as expected.

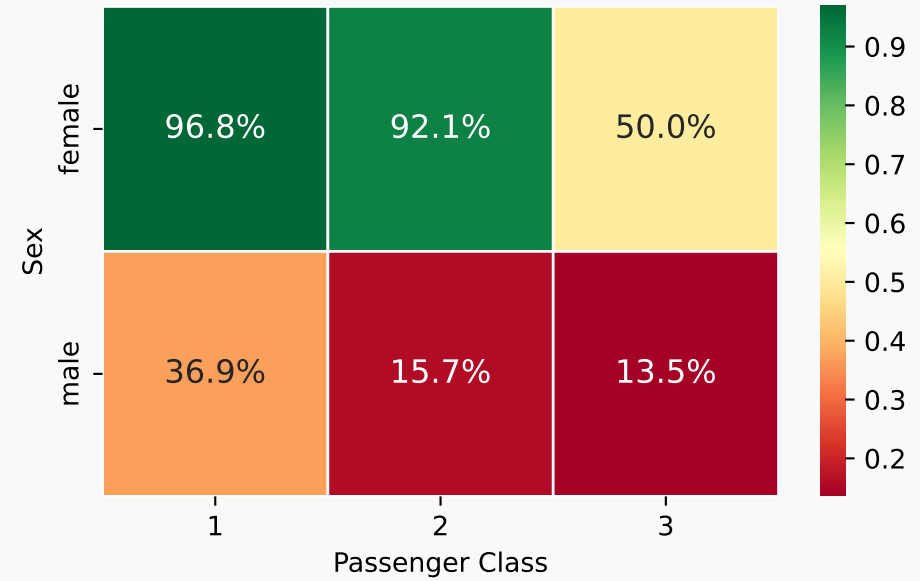
SECTION 5: Embarkation Port & Family Size

Where passengers boarded and how family structure affected survival

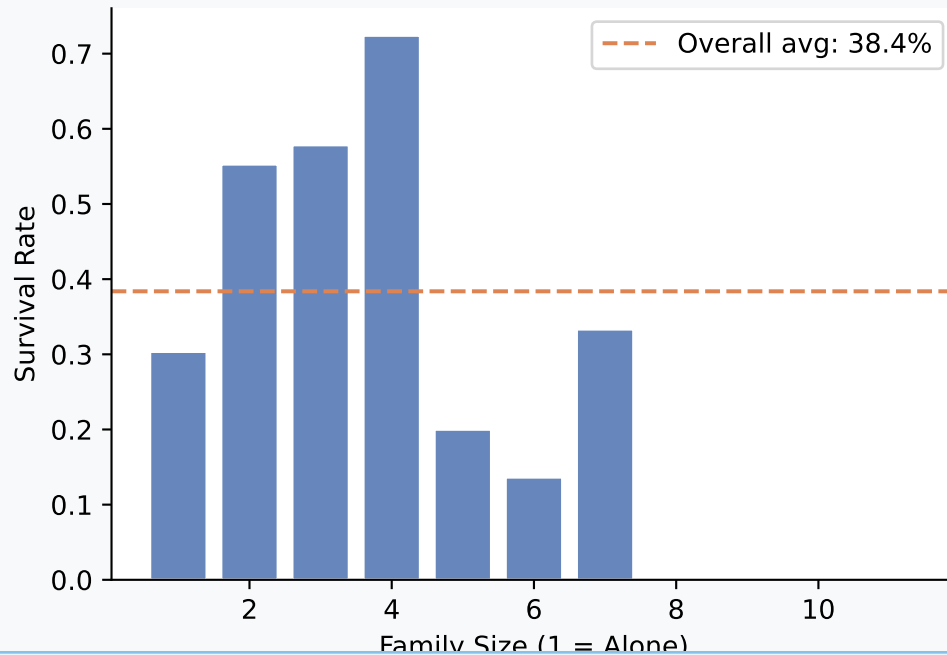
Survival Rate by Embarkation Port



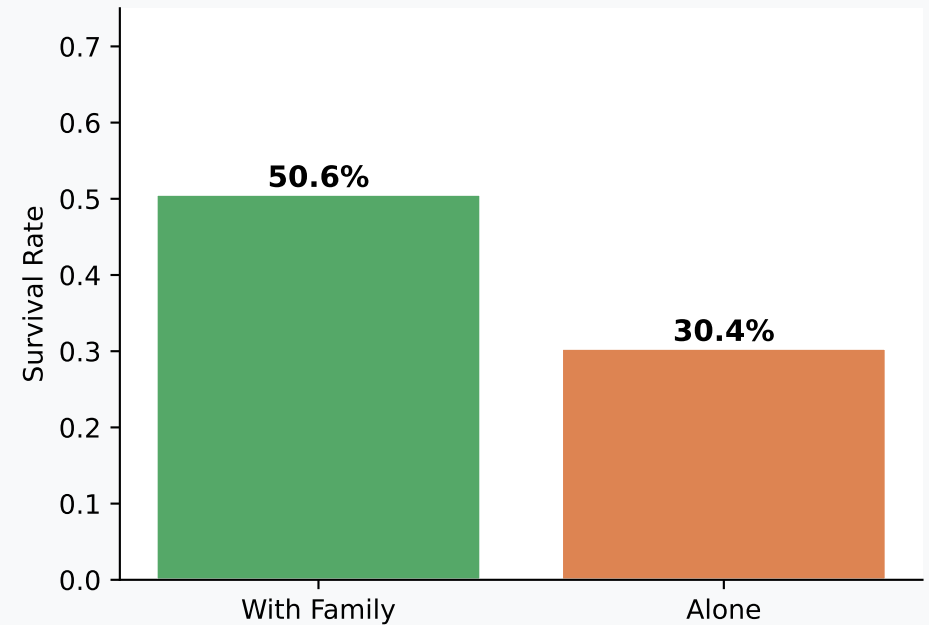
Survival Rate: Sex × Class



Survival Rate by Family Size



Survival: With Family vs Alone



Cherbourg's higher survival rate is explained by its higher proportion of 1st class passengers — a confounding variable. Small families (2-4 members) had the best survival outcomes.

SECTION 6: Summary of Key Findings

Consolidated insights from the Titanic EDA

#1 SEX IS THE #1 PREDICTOR

Female: 74.2% survived | Male: 18.9% survived

"Women and children first" protocol was strictly followed. A female was ~4x more likely to survive.

#2 PASSENGER CLASS REFLECTS INEQUALITY

1st Class: 63.0% | 2nd Class: 47.3% | 3rd Class: 24.2%

Wealthier passengers had cabins closer to decks, better lifeboat access, and more information.

#3 SEX × CLASS INTERACTION IS MOST POWERFUL

1st class females: 96.8% survived | 3rd class males: only 13.5% survived

The combined effect of gender and wealth determines survival almost completely.

#4 CHILDREN WERE PRIORITISED

Young children showed above-average survival rates. Age median is 28.

Age alone is a weak predictor but strengthens when combined with Sex/Class.

#5 FAMILY SIZE HAS A SWEET SPOT

Solo: 30.4% | Small families (2-4): 50-72% | Large families (5+): < 20%

Small groups could coordinate evacuation; solo travellers had no one to help.

#6 FARE & CLASS ARE STRONGLY LINKED

Fare is right-skewed: median £14 but mean £32 due to extreme 1st class prices.

Pclass-Fare correlation: -0.55. Higher fare → better survival odds.

#7 DATA QUALITY ISSUES

Cabin: 77% missing → drop or create binary "has cabin" flag.

Age: 20% missing → impute with median. Embarked: 2 missing → fill with mode (S).